

# MATLAB: Session 7

Vincenzo Antonio Isoldi  
Francesco Giuseppe De Vitis

13 November 2025

## Exercises

### Part A – Linear Applications and Change of Basis

In the following,  $\mathcal{E}_n = \{e_1, \dots, e_n\}$  denotes the canonical basis for  $\mathbb{R}^n$ .

**A.1** (*From last time*) Let  $h : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  be the endomorphism represented, with respect to the canonical basis  $\mathcal{E}_3$ , by the following matrix

$$C = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

**A.1.1** Assign the matrix  $C$  to the variable `C`.

**A.1.2** Check that the vectors

$$\begin{aligned} v_1 &= e_2 + e_3, \\ v_2 &= e_1 + e_3, \\ v_3 &= e_1, \end{aligned}$$

form a basis for  $\mathbb{R}^3$ .

**A.1.3** Let  $\mathcal{D}$  be the basis of the previous point. Compute  $M_{\mathcal{D},\mathcal{D}}(h)$ . Assign it to the variable `Mh_D`.

**A.1.4** Compute the dimension of  $\ker h$  and assign it to the variable `dimkerh`.

**A.1.5** Compute the dimension of  $\operatorname{Im} h$  and assign it to the variable `dimimh`.

**A.1.6** Compute a basis for  $\ker h$  and assign it to the variable `basiskerh`.

**A.1.7** Compute a basis for  $\operatorname{Im} h$  and assign it to the variable `basisimh`.

**A.1.8** Is  $h$  injective? Is  $h$  surjective?

## Part B – Eigenvalues, Eigenvectors, Diagonalizability

**B.1** Consider the endomorphism  $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  defined as

$$f(x, y, z) = (x, 2x - y + z, x + 5y + 3z).$$

- B.1.1** Assign to the variable **A** the matrix representing  $f$  with respect to the canonical basis of  $\mathbb{R}^3$ .
- B.1.2** Compute the characteristic polynomial of  $f$  and assign it to **charf**.
- B.1.3** Compute the eigenvalues of  $f$  and assign them to the column vector **eigf** (eigenvalues with algebraic multiplicity greater than one, must be repeated).
- B.1.4** Compute the eigenvectors of  $f$ . Save them as follows: if  $v_1, \dots, v_j$  form a basis for the eigenspace relative to the eigenvalue  $\lambda = a$ , then assign them as columns of the matrix **Vf\_a**.
- B.1.5** Is  $f$  diagonalizable? If yes, assign its diagonal form to the variable **Diagf**.
- B.1.6** If  $f$  is diagonalizable, compute the diagonalizing matrix and assign it to **P**.

**B.2** Consider the endomorphism  $g : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  defined as

$$g(x, y, z) = (x + 2y + z, -x + 4y + z, -x + y + 3z).$$

- B.2.1** Assign to the variable **B** the matrix representing  $g$  with respect to the canonical basis of  $\mathbb{R}^3$ .
- B.2.2** Compute the characteristic polynomial of  $g$  and assign it to **charg**.
- B.2.3** Compute the eigenvalues of  $g$  and assign them to the column vector **eigg** (eigenvalues with algebraic multiplicity greater than one, must be repeated).
- B.2.4** Compute the eigenvectors of  $g$ . Save them as follows: if  $v_1, \dots, v_j$  form a basis for the eigenspace relative to the eigenvalue  $\lambda = a$ , then assign them as columns of the matrix **Vg\_a**.
- B.2.5** Is  $g$  diagonalizable? If yes, assign its diagonal form to the variable **Diagg**.
- B.2.6** If  $g$  is diagonalizable, compute the diagonalizing matrix and assign it to **Q**.

**B.3** Consider the endomorphism  $h : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  defined as

$$h(x, y, z) = (z, 2x - x + y + z, -x + 2z).$$

- B.3.1** Assign to the variable **D** the matrix representing  $h$  with respect to the canonical basis of  $\mathbb{R}^3$ .
- B.3.2** Compute the characteristic polynomial of  $h$  and assign it to **charh**.
- B.3.3** Compute the eigenvalues of  $h$  and assign them to the column vector **eigh** (eigenvalues with algebraic multiplicity greater than one, must be repeated).
- B.3.4** Compute the eigenvectors of  $h$ . Save them as follows: if  $v_1, \dots, v_j$  form a basis for the eigenspace relative to the eigenvalue  $\lambda = a$ , then assign them as columns of the matrix **Vh\_a**.
- B.3.5** Is  $h$  diagonalizable? If yes, assign its diagonal form to the variable **Diagh**.
- B.3.6** If  $h$  is diagonalizable, compute the diagonalizing matrix and assign it to **R**.